Status	Finished
Started	Tuesday, 20 October 2026, 11:25 AM
Completed	Tuesday, 20 October 2026, 11:29 AM
Duration	3 mins 43 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Combining sorted sublists
- c. Sorting the entire list sequentially
- d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Inserting data into a list
- b. Arranging data in a particular format
- c. Deleting data from a list
- d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are discarded
- c. They are compared element by element
- d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **4**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 80,150,200,470,40,90,300,400,70,120
- c. 200,470,80,150,40,90,300,400,70,120
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It compares adjacent elements and swaps them if necessary
- b. It builds a sorted array one element at a time
- c. It repeatedly finds the minimum element and moves it to the sorted part of the list
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **6**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging
b. Searching
c. Complexity
d. Sorting

The correct answer is:
Complexity

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Bubble Sort
b. Heap Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. Searching
c. Selection
d. All of the above

The correct answer is: All of the above

Question **9**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Heap sort

b. Bubble sort

c. Quick sort

d. Insertion sort

The correct answer is:
Insertion sort

Question **10**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are ignored

b. Each part is searched for a specific element

c. Each part is sorted independently

d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays

b. It sorts data out of the box efficiently

c. It requires external libraries

d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **12**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It allows for quicker identification of duplicates
c. It complicates the analysis process
d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **13**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method
b. Divide and conquer
c. Dynamic programming
d. Backtracking

The correct answer is:
Divide and conquer

Question **14**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data sequentially
b. Rearranging data without sorting
c. Dividing the input into parts, solving each part, and combining the solutions
d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Merging two datasets
- c. Finding a specific element in a list
- d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order